

Aayush Pokhrel

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PROFESSIONAL SUMMARY

A Computer Science graduate from the University of Cincinnati with hands-on experience in machine learning, backend engineering, and full-stack data applications. Built and deployed multi-service ML and data products end to end using Python, FastAPI, PostgreSQL, TensorFlow, and PyTorch, including from scratch transformer work. Seeking an ML or software engineering role with real ownership and room to learn from senior engineers.

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, C++, C#, Java, MATLAB, HTML, CSS

Frameworks/Libraries: FastAPI, Node.js, React, Next.js, React Native (Expo), TensorFlow, PyTorch, XGBoost, NumPy, SciPy, Tailwind CSS

Data/DevOps: PostgreSQL, Docker, Docker Compose, GitHub Actions (CI/CD), REST APIs, Power BI, Tableau, Wireshark

Software/OS: Adobe CC, Microsoft Office; macOS, Windows, Linux

EDUCATION

University of Cincinnati

April 2026

Bachelor of Science, Computer Science

GPA: 3.45 | Dean's List (5 terms)

Relevant coursework: Deep Learning (from-scratch LLMs), Software Engineering, Algorithms, Operating Systems

WORK EXPERIENCE

Autonomous Systems Research Intern

Jan 2025 – Aug 2025

University of Dayton Research Institute

- Researched DNA structure as a candidate for quantum computing architectures, solving the Schrodinger equation in Python (NumPy/SciPy, QuTiP, Qiskit) to quantify information storage and transfer versus silicon systems
- Cut simulation runtime ~90% (~48h to ~5h) across 100+ runs via numerical vectorization, an improved integration method, GPU acceleration, and parallelization
- Built PyTorch surrogate models to approximate and optimize the quantum simulation dynamics, and applied MATLAB/Simulink to modeling, simulation, and testing of autonomous vehicles and control systems

Project Management Co-op

Aug 2023 – Dec 2023

EMCOR Facilities Services

- Collaborated with cross-functional project management teams to support the full project lifecycle, from proposal development to project close-out documentation
- Built and maintained Power BI and Tableau dashboards using data from Smartsheet to visualize KPIs, monitor project performance, and support data-driven decision-making
- Managed and maintained documentation and reporting for 120+ active projects, ensuring accurate tracking of timelines, deliverables, and issues across stakeholders

PROJECTS

Playstat: Sports Analytics Dashboard with Real-Time Data Integration

Aug 2024 – Present

- Built a FastAPI/PostgreSQL backend and Next.js dashboard ingesting ~1M stat rows across 6 leagues from 4 external APIs via 3 scheduled daily jobs
- Developed an XGBoost model with negative-binomial distributions and conformal calibration; retired it after backtesting underperformed market baselines, now rebuilding on better data
- Designed a combinatorial optimizer ranking outcomes by de-vigged joint probability, backed by a 98-test pytest suite and GitHub Actions CI

Budgerr: Personal Finance and Budgeting App

Aug 2025 – Present

- Built a full-stack budgeting app with a FastAPI/PostgreSQL backend serving React Native (Expo) mobile and Next.js/TypeScript web clients, using the Plaid API for automated transaction sync
- Implemented an envelope-budgeting engine with category limits, threshold alerts, and trend analysis, plus recurring-charge and price-hike detection via custom clustering logic
- Deployed with Docker Compose, API-key auth, encrypted nightly backups, and GitHub Actions CI across all three repos (~78 tests)

AI Inventory Management System

Aug 2025 – Apr 2026

- Enabled real-time item tracking by integrating a TensorFlow image-recognition model with a Node.js REST API and PostgreSQL schema (with user auth)
- Built the integration layer for the model, then ran system-wide integration and performance testing across subsystems
- Built a real-time warehouse simulation modeling item flow through the recognition pipeline to test tracking accuracy and latency at scale